Nasser Mohammed

Department of Astronomy and Astrophysics
University of Toronto

✉: nasser.mohammed@astro.utoronto.ca
🔗: astro.utoronto.ca/~nasser.mohammed
iD: 0009-0008-1224-0382

Research Interests: Galactic Archaeology, Near-Field Cosmology, Astrostatistics, Machine Learning

Education

Sep. 2024 – Present **PhD Candidate in Astronomy & Astrophysics | University of Toronto**
Thesis topic: *Indirect Dark Matter Detection via Stellar Streams*
Supervisor: Professor Ting S. Li

Sep. 2019 – Dec. 2023 **BSc Honours in Physics | University of British Columbia, Okanagan**
Minor in Data Science
Thesis Title: *Polarimetric Study of 'Tadpole' Feature G137+7 with CHIME Radio Data*
Supervisor: Professor Alex S. Hill

Publications

First Author Publications

N. Mohammed, et al. 2026, ApJ, 1005, 2: *The Kinematically Hot, Extremely Metal-Poor C-19 Stellar Stream...*
N. Mohammed, et al. 2024, ApJ, 971, 100: *Faraday Tomography with CHIME: The 'Tadpole' Feature G137+7*

Co-Authored Publications

R. Carlberg, T.S. Li, E. Jarvis, **N. Mohammed**, et al. 2026, submitted to ApJ: *Joint Modeling of GD-1 and C-19 as Old Streams*
E. Jarvis et al. (incl. **N. Mohammed**) 2026, submitted to ApJ: *Characterizing the GD-1 Stream with DESI DR2 Data: Thin Stream and Hot Cocoon*
A. Byström, et al. (incl. **N. Mohammed**) 2026, submitted to MNRAS: *Constraining the Galactic bar using the M92 stellar stream*

Awards & Honours

2026-2029	*$120,000*	NSERC Canadian Graduate Research Scholarship - Doctoral (CGSR D)
2025-2026	*$27,000*	NSERC Canadian Graduate Scholarship - Masters (CGS M)
2025-2028	-	LSST Discovery Alliance Data Science Fellowship
2025	*$10,000*	UofT Recognition of Excellence Award
2024	*$7,000*	Mary Louise & Ronald Laidlaw Martin Graduate Scholarship in A&A
2024	*$5,000*	David A. Dunlap Department of Astronomy & Astrophysics Entrance Fellowship
2024	*$10,000*	NSERC Undergraduate Student Research Award
2023	*$11,000*	Irving K. Barber Faculty of Science Undergraduate Research Award
2023	*$10,000*	NSERC Undergraduate Student Research Award

Research Supervision

May 2026 – Aug. 2026	Emily Moran	Research Project Primary Supervisor | Mitacs
May 2025 – Sept. 2025	Alejandra Calderon	Research Project Co-Supervisor | Mitacs

Further Education & Select Attendances

Feb. 2025 – Present LSST Discovery Alliance Data Science Fellowship Program
6 week-long workshops on select data science/AI topics for astronomers.

Feb. 2026 RE:SciComm - Canada's Science Communication Conference
May 2023 Canadian Astroparticle Physics Summer School, Queen's University & SNOLAB

Leadership & Service

Jul. 2026	Local Organizing Committee Member: "Near Field Cosmology in the Era of Big Data: Local Group and Beyond" Workshop | The Fields Institute, University of Toronto
Oct. 2025 – Current	Union Steward | CUPE 3902 Unit 1 Union Steward for Unit 1 & Graduate Student Astronomy Association Union Representative.
Oct. 2024 – Current	Colloquium Speaker Dinner Organizer | David. A Dunlap Dept. of Astro. & Astrophysics Co-ordinates post-doc and graduate student attendees, place reservations, accomodate speaker and attendee dietary restrictions.
Oct. 2024 – Oct. 2025	Mediation Committee | Graduate Astronomy Student Association Authored report on graduate student population financial wellbeing and work-life balance. Successfully advocated for increase in astronomy graduate student stipends.
May 2020 – May 2024	Co-Founder & President of the UBC Okanagan Astronomy Club Hosted seminars, coordinated stargazing nights, organized youth STEM outreach initiatives.

Media Appearances

June 2025	*Telescope images give 1st ultra HD images of the universe* | CBC News On-camera interview; clip included in aired segment

Observing Experience

26 Sep. 2025 – 26 Sep. 2025	Co-Investigator, Las Campanas Observatory | Región de Coquimbo, Chile 2 nights on Magellan/MIKE; Observations of the C-19 stellar stream

List of Talks & Lectures

July 2026	Near Field Cosmology in the Era of Big Data | University of Toronto Contributed Talk
May 2026	AST101 Substitute Lecturer | University of Toronto 2 hr Lecture
Oct. 2025	IAU Symposium 403: Hidden Beauty of the Galactic Outskirts | Córdoba, Spain Contributed Talk
Oct. 2025	AST101 Substitute Lecturer | University of Toronto 1 hr Lecture
Aug. 2025	York University EXPLORE Summer School | North York, ON, Canada Lecturer + Tutorial Leader
Jun. 2025	Clusters in Halifax Workshop | Halifax, NS, Canada Contributed talk
Jun. 2025	CASCA Annual General Meeting 2025 | Halifax, NS, Canada Poster presentation
Mar. 2024	Okanagan College | Kelowna, BC, Canada Guest lecture: Introduction to the Interstellar Medium (ASTR 121)
Feb. 2024	University of Calgary PHAS Symposium | Calgary, AB, Canada Contributed talk
Sep. 2023	UBC Okanagan Faculty of Science URA Symposium | Kelowna, BC, Canada Poster presentation
Jun. 2023	CASCA Annual General Meeting 2023 | Penticton, BC, Canada Poster presentation
Jun. 2023	TomFest: Frontiers in Radio Science and Engineering | Kaleden, BC, Canada Contributed talk